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Multi-Timescale Control of Smart Inverters for Optimal Operation of Low-Inertia Grids

Himanshu Grover¹  | Sumedha Sharma² | Ashu Verma³ | Innocent Kamwa¹

¹Department of Electrical and Computer Engineering, Laval University, Québec City, Quebec, Canada | ²Department of Electrical and Software Engineering, University of Calgary, Calgary, Alberta, Canada | ³Department of Energy Science and Engineering, Indian Institute of Technology Delhi, New Delhi, India

Correspondence: Himanshu Grover (groverhimanshu.ee@gmail.com)

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ABSTRACT

This article proposes a novel frequency and voltage control scheme for low-inertia electrical systems with high penetration of renewable energy sources (RES). A multi-timescale coordinated control scheme was proposed to optimally control inverter-based resources in different timescales. Accordingly, a two-stage stochastic optimisation framework has been developed for optimal operation of battery energy storage system (BESS) and voltage source converters (VSC) in hour-ahead and intra-hourly timescales, to counteract the effects of uncertainties in solar photovoltaic (PV) and load. Additionally, a novel real-time coordination framework was developed for fast frequency control, triggered by appliance switching/scheduling information through energy internet. Thus, real-time control is implemented as a pre-disturbance preventive action, appropriately acting with the load switching event. Furthermore, the proposed real-time frequency control is developed as a coordination strategy for primary regulation by adaptive VSC control and recovery control by the grid. Extensive simulations were performed to verify suitability of the proposed optimisation and control strategy in mitigating the effects of unforeseen uncertainties and scheduled events on system stability. Effectiveness of the proposed control is further verified by experimental validation on laboratory-scale hardware test setup.

1 | Introduction

Motivated by increasing concerns over carbon emissions and climate change caused by excessive use of fossil fuels, the global energy sector is transitioning towards integration of clean energy sources such as solar PV, wind, etc. In view of this, countries around the world have set ambitious targets for grid penetration of RES. However, the proliferation of RES in distribution systems has a detrimental effect on system inertia. The integration of distributed renewable energy sources (RES) on a wide scale leads to a decrease in the inertia of the grid. This decrease in inertia produces significant fluctuations in the frequency and voltage of the system because of power imbalances. The increasing popularity of distributed photovoltaic (PV)

systems has heightened the significance of frequency regulation. Figure 1 depicts a comparison of the frequency response to a variation in system power between a traditional power grid and a grid that incorporates renewable energy sources [1]. It is noteworthy that system inertia is critical for maintaining stability of power systems during disturbances by limiting the rate of change of frequency (RoCoF). Consequently, displacement of power generation from synchronous generators to power electronics based RES generators creates issues for the electrical microgrid frequency and voltage stability.

Power system frequency and voltage must be maintained within safe operating limits during steady state and dynamic conditions, to avoid undesirable operation of protection relays and

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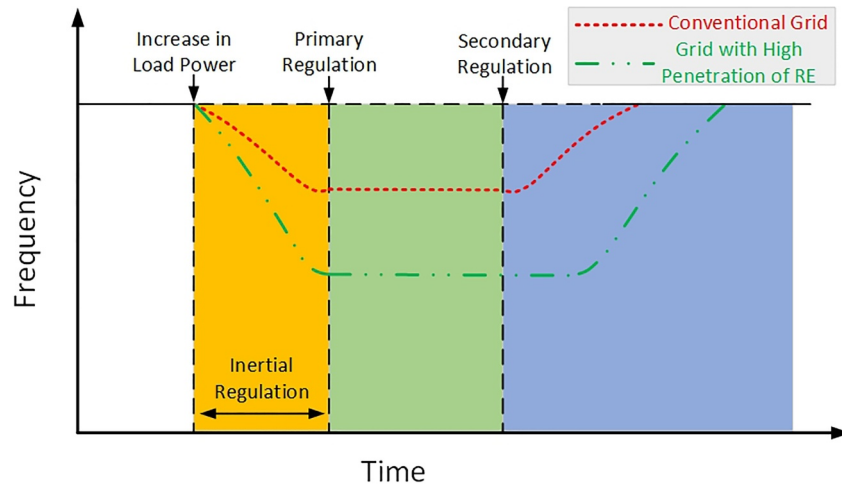


FIGURE 1 | A comparison of the frequency response to a variation in system power.

disconnection of generators [2]. However, widespread integration of RES is bound to cause a reduction in system inertia and the introduction of power uncertainties in the system, which compromises the effectiveness of traditional under-frequency load shedding (ULFS) [3] and ROCOF schemes. In view of this, recent research works have investigated control strategies to counteract the stability concerns arising in high RES penetrated electrical networks through battery energy storage systems (BESS) [4, 5], PV inverters [6–11], event-triggered [12] and load control [13, 14].

The authors of reference [4] presented an active distribution system for voltage regulation and line congestion management through BESS and PV, whereas the authors of reference [5] utilised BESS for frequency support in microgrids. Further, the works presented in references [6, 7] developed hierarchical voltage control through PV inverters. Other works including the works in references [8, 9] investigated frequency regulation through inverter-based PV systems. The authors of reference [10] presented an active-reactive power control strategy for grid-tied PV inverters. Moreover, the work presented in reference [11] developed an active power control scheme of PV system to support inertial response of generators and frequency control.

Apart from these, several papers investigated frequency and voltage stability problems for particularly low-inertia networks [2, 15–17]. The authors of reference [15] presented a study of ULFS protection scheme for a low-inertia scenario of the Great Britain power system with rising RES penetration. The authors of reference [16] developed the concepts of frequency inertia and voltage inertia, and a distributed algorithm to limit RoCoF and rate of change of voltage. Similarly, the authors of reference [17] proposed a frequency control strategy for low-inertia systems through synchronous condensers and synthetic inertia of wind power plants. It is noteworthy that most existing works focus on controller design for real-time implementation alone, and do not focus upon optimal operation of resources which is essential for overall long-term system performance.

However, although the works in references [6, 7] presented optimal operational planning models, the task of optimal dispatch is concerned with offline dispatch decisions for energy

management and does not focus upon real-time voltage and frequency regulation. It is noteworthy that time-ahead optimal dispatch decisions are not well suited for optimal real-time control since uncertainties render the time-ahead decisions sub-optimal in real time. This results in large deviations and steady-state errors in real-time voltage and frequency regulation. This envisages the need for developing real-time optimal decision-making and control. Even though a real-time droop control for PV inverters was developed in reference [18], this work does not demonstrate practical performance during actual system operation. Moreover, these research works have developed only voltage/reactive power control, whereas frequency control was not focused upon.

In reference [19], two voltage control algorithms were suggested as plug-and-play solutions that effectively minimise power distribution losses. The paper [20] introduced an enhanced droop control technique that carries out two crucial functions. The first function involves reducing errors using low-bandwidth synchronisation signals. The second function involves recovering the voltage by utilising the output voltage. In reference [21], a power-sharing tiny signal converter was proposed as an adaptive droop control mechanism. This approach does not employ a voltage gradient inside its construction and utilises an inverter to regulate the output voltage. The paper introduced a self-optimised control method that utilises power-frequency droop control, as demonstrated in [22]. The frequency-active power-droop control can function as a hierarchical control. The adaptive current-voltage droop control, presented in reference [23], utilised a photovoltaic (PV) battery and exhibited excellent performance for power provision in developing regions. By employing the model predictive control (MPC) methodology and integrating it with the voltage-current ($V-I$) droop method, the power quality was enhanced and DER were safeguarded from overloading [24]. MQTT, as proposed in reference [25], enhanced the main droop control and was applied in several settings.

Various advanced schemes were included in the literature to provide frequency regulation using real-time control of inverters such as like linear-quadratic regulator (LQR) [26], model predictive control (MPC) [27, 28], nonlinear control [29] and data-driven based approaches such as ANN [30], deep reinforcement

learning (DRL) [31, 32] and recurrent neural network (RNN)-based RL [33]. In order to monitor and share the frequency deviation and RoCoF and so coordinately optimise the hydro-governor's action, a simulated annealing algorithm (SAA) was devised [34]. To improve the primary frequency response, a related concept based on wide area RoCoF sharing schemes was also proposed in reference [35]. However, such algorithms increase complexity in the system by providing more computational burden to the central which not only increases computation time but also increases overall system operating cost.

It is further worth mentioning that most real-time controllers for load frequency and bus voltage regulation perform corrective measures post-disturbance. This implies that the control action is triggered whenever the frequency/voltages cross threshold limits of nominal operation. Whereas post-disturbance corrective action will recover the system frequency/voltages to its nominal state, it must be noted that low-inertia networks witness significant voltage and frequency deviations even for small disturbances such as load switching and for intermittency in solar PV output. Such disturbances lead to high RoCoF, resulting in undesirable triggering of the protection system. However, because in a smart grid environment, load switching and scheduling decisions are known a priori with a lead-time of few seconds to few hours depending upon the control and scheduling process, network voltage and frequency fluctuations maybe readily avoided through pre-switching control rather than post-disturbance corrections. To the best of the authors' knowledge, existing works do not focus on preventive control by making use of scheduling and control information.

In view of the aforementioned discussion, this paper develops a multi-timescale coordinated control framework for adaptive control of VSCs in low-inertia electrical systems. Major contribution in this context includes novel primary frequency and voltage regulation by optimal power dispatch of solar PV and BESS based VSCs. Energy and power management strategy is distributed over multiple time-scale through two-stage stochastic programming framework for hour-ahead and intra-hourly optimisation considering PV and load uncertainties. VSC operation is re-dispatched to provide impulse power through event-triggered real-time optimisation based on information of appliance operation through energy internet. Furthermore, a recovery control strategy is developed to restore the active and reactive power set points of VSC to their initial state while shifting incremental load power to the grid. Major contributions are summarised as follows.

1. Development of a novel primary frequency and voltage control strategy through optimal coordination and operation of multiple VSCs in smart distribution networks. VSC power dispatch is optimised at multiple timescale in response to uncertainties and load switching information in an energy internet paradigm.
2. Multi-timescale management of energy and power through adaptive control of VSCs to mitigate the effects of short lead-time uncertainties and large real-time load switching events by developing a real-time two-stage stochastic, and real-time optimisation and control framework.

3. Real-time optimisation and coordination control framework for an adaptive VSC control to provide impulse load power followed by recovery control method. Additionally, to restore optimal dispatch set-points of generation resources following primary frequency regulation.
4. Practical deployment and performance verification of the developed optimisation and control strategy on laboratory-scale experimental test setup.

It is worth highlighting that multi-level control of microgrids as discussed in literature comprises of similar timescales and are commonly referred to as primary, secondary and tertiary controls. However, the specific challenge and the prime area of interest for research is the problem in integrating the models developed in programming and simulation to physical control devices on a system level. This is a challenge for industrial application of the multi-level control that is discussed commonly in literature.

Integrating programming, simulation and physical control—especially across multiple timescales—is a recognised challenge in engineering and cyber-physical systems. This complexity arises because each domain (software, simulation and hardware) typically operates with its own models, interfaces and timing requirements, making seamless interoperability difficult. For example, simulation tools are often domain-specific and vertically integrated, whereas real-world control requires horizontal integration across these domains, ensuring that data and control signals are synchronised and semantically consistent.

Whereas hour-ahead and intra-hour optimisation are short-term operational planning activities and do not necessarily demand real-time communication and control, the real-time stage incorporates physical control, which is particularly challenging to achieve. In order to address this existing challenge, we developed a real-time optimisation, control and communication framework that also integrates the peculiarities of physical power networks, device-level characteristics of VSCs and practical challenges of communication channels.

2 | System Modelling

2.1 | Overview of Proposed System

The smart grid paradigm envisions an intelligent, dynamic and highly interconnected energy network that optimises the generation, distribution and consumption of energy through advanced digital technologies. Overview of the proposed system integrated in a distribution network is shown in Figure 2. The integration of loads, resources such as solar photovoltaic (PV) systems, battery energy storage systems (BESS) and other energy resources into the power networks is enabled by the energy internet through the development of a smart power infrastructure. This infrastructure utilises enhanced metring, sensors and communication networks to facilitate the instantaneous exchange of data and control. The building operator and central controller constitute the distributed energy management and control system which maximise the equilibrium between the

availability and consumption of energy, while intelligent appliances and Internet of things (IoT) devices facilitate the ability to respond to demand and utilise energy efficiently. PV and BESS are controlled integrated and controlled through suitable power electronic converters, energy management systems, optimisation and analytical tools. This integration ensures a robust and efficient energy internet ecosystem. As depicted in the figure, multiple buildings at the low-voltage (LV) distribution level are connected to the utility grid through a sub-station. Each building is equipped with smart metring infrastructure, controllable loads and distributed energy resources (DER) such as solar PV and BESS. The smart metre measures the amount of net-power exchanged by the building with the LV network. The measured data from the smart metre is wirelessly transmitted and stored in a cloud database using Internet-of-things (IoT) based energy internet and monitored by central controller.

A more elaborative description of the proposed system for an individual building is illustrated in Figure 3. It can be observed from the figure that the building is connected to the utility grid through a smart metre at the point-of-common coupling (PCC). A rooftop solar PV system and BESS are connected to the

building's DC bus via DC-DC boost converter and DC-DC bi-directional converter, respectively. The DC bus is interfaced to the AC bus through a VSC, which is interchangeably referred to as the smart inverter hereon. Further, the building loads are connected to the AC bus through a smart load control board, which receives load control signals from a building operator through WiFi. Additionally, the power electronic converters include a WiFi-based bi-directional communication feature that allows them to communicate with the central controller.

The building operator and central controller communicate with the cloud database over multiple timescales for assistance and implementation of their control decisions. The building operator coordinates the real-time connection or disconnection of building loads by sending the control command over the cloud database, utilising a WiFi-based bi-directional communication function. Furthermore, the central controller is equipped with a high-computational micro-controller that gathers information from the cloud database and performs multi-timescale optimisation to achieve optimal dispatch signals at various timescales. The multi-timescale optimisation is focused on the optimal load frequency and voltage control requirements of the network,

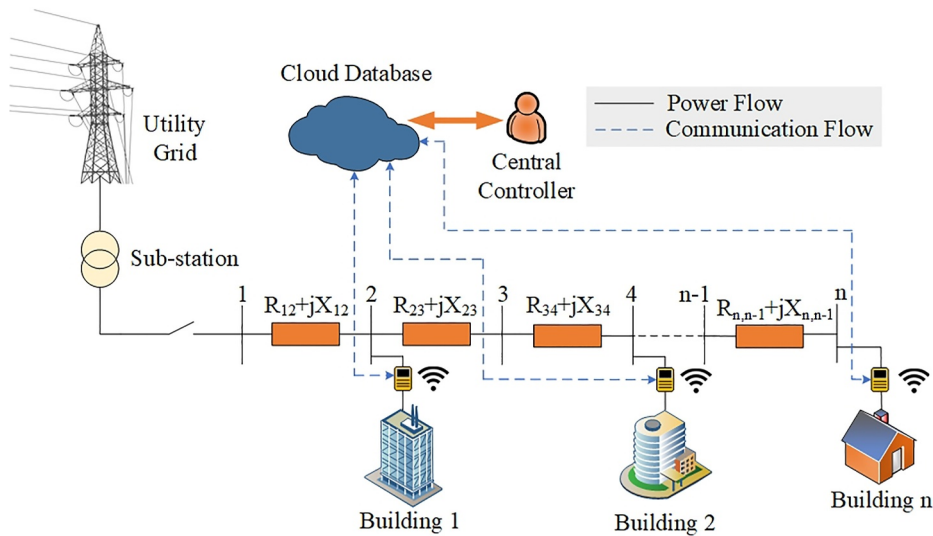


FIGURE 2 | Overview of the proposed system.

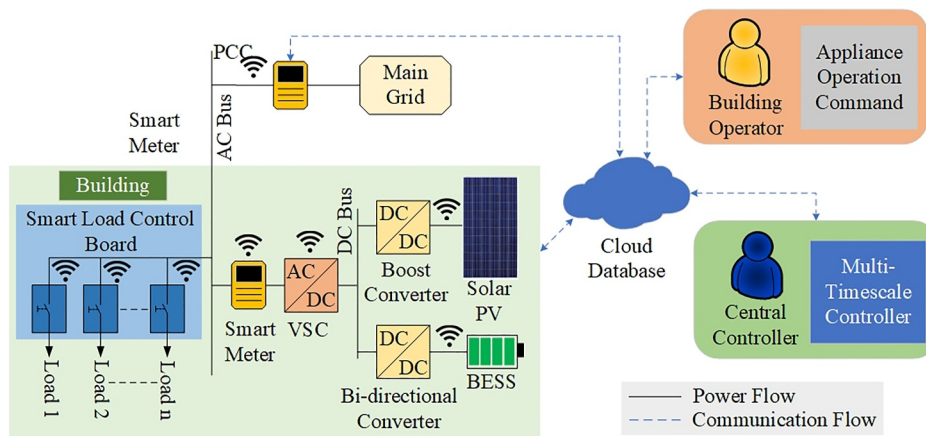


FIGURE 3 | Detailed description of the proposed system for a building.

thereby, the optimum dispatch signals are communicated to individual local controllers via distributed control approach.

2.2 | Multi-Timescale Coordinated Control Strategy

A multi-timescale coordinated control strategy was proposed to maintain real-time active and reactive power generation balance for RES integrated electrical microgrid systems. The term ‘multi-timescale’ essentially refers to the multiple distinct timescales at which the proposed optimisation and control are performed. Owing to the occurrence of uncertainties and switching events during actual system operation, the proposed control has been developed at three stages operating in three different timescales. The multi-timescale approach includes hourly, intra-hour and real-time optimal power control of system resources, that is, VSC and BESS. The hour-ahead operation is performed every hour, the intra-hour optimisation is performed at every 15 min and the real-time optimisation is an event-triggered optimisation that is performed as and when the central controller delegates a large switching decision in practical real time. The framework of the multi-timescale coordinated control strategy is illustrated in Figure 4.

At the first stage, an hour-ahead coordination method is designed. The hour-ahead stage corresponds to hourly active and reactive power dispatch decisions of BESS–VSC, based upon RES and load forecast information as input data, considering uncertainties of solar PV and load. This mode performs optimal dispatch through the solution of an optimal power flow model for the network, which ensures optimal system operation. Although the hour-ahead decisions of VSC and BESS power dispatch, and power drawn from the grid are optimal for

deterministic conditions of the network, the control outputs tend to be sub-optimal during actual system operation because of the occurrence of uncertainties in PV output and load in shorter timescales. Consequently, the grid would witness the RES-penetrated distribution networks as highly variable net-loads with low observability along the network, which would further affect grid operations.

In order to minimise the impact of uncertainties occurring during smaller timescales on grid operations, and to ensure optimal operation of the RES-penetrated distribution networks which are prone to higher volumes of reverse power flows and over voltages, the intra-hour optimisation problem was developed at the 15-min timescale. Thus, similar to the hour-ahead stage, the intra-hourly stage corresponds to optimal re-dispatch of VSC and BESS power to counteract the effect of uncertainties in RES and load occurring during actual system operation. Accordingly, in the 15-min timescale, the central controller is provided with updated uncertainty forecasts and network information (line flows and voltages) to control any adverse effects on network operation through re-dispatch of VSC and BESS.

It must be noted that the hour-ahead and intra-hourly optimisation stages constitute a short-term operational planning activity that is performed to optimise the overall system operation. However, because microgrids with large RES penetration have reduced inertia, large real-time disturbances may affect the stability of the system. Consequently, the third stage of control mechanism is required to control the voltage and frequency of the system during real-time events of large disturbances, occurring within the 15-min intervals. It is noteworthy that primary frequency control by the grid would result in a higher RoCoF, large frequency deviation and thus, time delay because

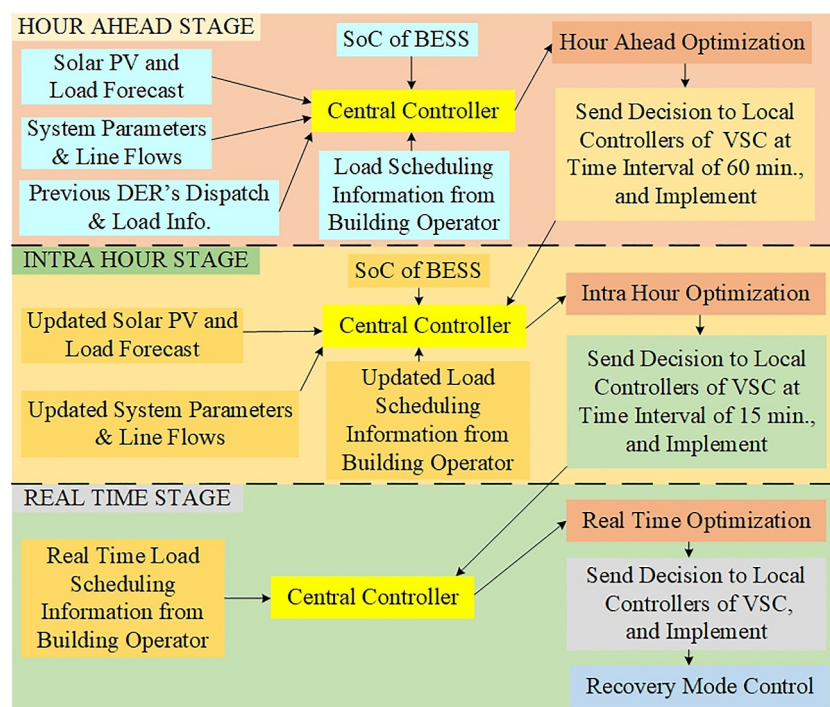


FIGURE 4 | Multi-timescale coordinated control framework.

of the mechanical dynamics of the grid, which can be suitably mitigated by real-time control of fast-acting power electronic converters. Accordingly, the central controller performs real-time optimisation for primary frequency and voltage regulation through VSC at the third stage of the proposed control framework. After the primary frequency regulation is achieved in real-time, a recovery-control method is implemented to stably restore the active-reactive power dispatch of VSC to their initial operating states and gradually shift the increase in power demanded by the real-time disturbance onto the grid in the longer timescale without causing any frequency deviation. The proposed methodology is discussed in detail in the successive sections.

3 | Mathematical Modelling

3.1 | Inverter Capability

To incorporate the active-reactive power support through inverter, it is necessary to understand the capability limits of an inverter. Generally, the PV inverters are designed with 25% overrated capacity of their active power and the underutilised capacity can help in reactive power support for voltage regulation. For the analysis of P - Q capability of an inverter, a simplified model is used and the phase diagram of grid connected inverter is shown in Figure 5 [36]. The equations for active and reactive power output of VSC are given as follows:

$$P = \frac{3V_g V_c \sin(\delta)}{X}; Q = \frac{3V_g}{X} (V_c \cos(\delta) - V_g). \quad (1)$$

Solving the above equations, the new equation is obtained as follows:

$$P^2 + \left[Q + \frac{3V_g^2}{X} \right]^2 = \left[\frac{3V_g V_c}{X} \right]^2, \quad (2)$$

where P is the oversized active power capacity of inverter. Q is reactive power of inverter, V_g is bus voltage across grid and V_c is bus voltage across inverter. X is the equivalent line reactance formed by filter inductance (L_f) and line inductance (L_g). I_g is current generated by inverter at an angle θ with grid voltage and δ is the angle between grid and converter voltages, Equation (3) represents the equation of circle which has centre position $\frac{-3V_g^2}{X}$ on reactive power axis and the radius of circle is given by $\frac{3V_g V_c}{X}$.

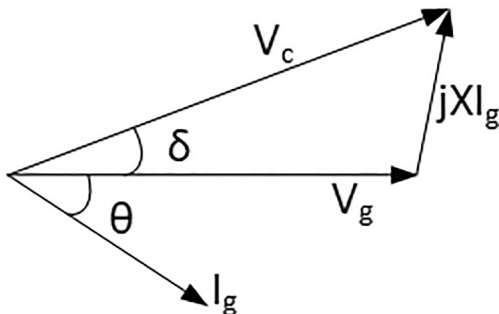


FIGURE 5 | Phasor diagram of grid connected inverter.

The reactive power limit at any instant of time can be calculated as follows:

$$Q = \sqrt{\left(\frac{3V_g V_c}{X} \right)^2 - P^2} - \frac{3V_g^2}{X}. \quad (3)$$

The reactive power injection capability of an inverter depends on the grid voltage of the associated bus. More reactive power can be injected with lower bus voltage and vice versa. The inverter reactive power capability model has been included in the multi-timescale for optimal optimisation dispatch of VSC. Inclusion of inverter reactive power capability to optimisation problem provides faster voltage regulation through optimal power dispatch of VSC.

3.2 | Optimal Operation Framework

To ensure optimal performance of the proposed system, control decisions of the system components must also be motivated at short-term operational objectives, in addition to obtaining suitable real-time performance. Accordingly, an optimal operation model was developed as a multi-stage coordinated optimisation problem as illustrated in Figure 4. Initially, short-term operational planning is performed in hour-ahead and intra-hourly (15-min) timescales to dispatch system components according to their response times and mitigate the effects of short-term uncertainty occurrence. This is followed by real-time optimisation and control through the smart inverter in real-time to counteract the effects of sudden load changes on system frequency.

Because network optimisation is computationally expensive and impractical to solve and implement in real-time, an operational planning problem was developed to dispatch system components time-ahead, while taking into consideration the effects of unforeseen uncertainties arising during actual system operation. Accordingly, a two-stage stochastic programming model was developed which time-decomposes an optimal power flow model for the short-term proposed operational planning problem using smart inverter based control in an RES-penetrated microgrid. The first stage optimisation performs a cost minimisation problem that determines optimal decisions of procuring electricity from the low-inertia grid and from BESS. The importance of the first stage optimisation lies in the fact that the decisions regarding grid energy purchase are enabled by the availability of grid retail tariffs which are discrete signals and broadcast hourly. This enables computation and implementation of the grid energy purchase decisions by the microgrid or building as a large consumer. The second stage optimisation is essential to perform a network re-optimisation upon the occurrence of uncertainties in the network which are unknown in longer hourly timescales. Accordingly, at the second stage a recourse optimisation is performed where the objective is loss minimisation, and the batteries are re-dispatched to mitigate the uncertainty occurrence.

Accordingly, the optimal power flow model is sequentially solved through coordination between the hourly and intra-hourly optimisation models. Appropriately, decisions regarding grid

energy consumption are computed and implemented every hour because of the hourly availability of grid tariffs, whereas owing to smaller response times, BESS and VSC power are dispatched every 15-min as and when updated forecasts are available. The objective of the optimal power flow model for short-term operational planning is minimisation of operational cost and grid energy consumption, represented mathematically as follows.

$$\min \left\{ C_g P_g + \sum_{j \in N_b} \sum_{h \in N_{ph}} C_b \left[P_{d,j,h}^b \eta_b^d + P_{c,j,h}^b / \eta_b^c \right] \right\} \Delta t, \quad (4)$$

where C_g and C_b refer to per unit grid energy cost and BESS operational cost, respectively. P_g and $P_{d/c}^b$ refer to the power drawn from the grid and discharging/charging power of BESS, respectively. N_b and N_{ph} refer to the sets of buses and phases, respectively. $\eta_b^{d/c}$ refers to the discharging/charging efficiencies of the battery. The objective presented in Equation (4) is computed hour-ahead considering the absence of uncertainties, whereas during intra-hourly operation, a re-dispatch optimisation is performed to counteract the effect of uncertainties occurring during actual system operation. The intra-hourly optimisation focuses on re-dispatching the VSC and BESS power to mitigate the effects of uncertainties on the network operation while minimising line losses because of increased bidirectional flows in the network. The second stage optimisation objective is therefore represented as follows.

$$\min \sum_{\substack{h \in N_{ph} \\ i \in I(j)}} r_{ij,h} \Delta I_{ij,h}^2 \Delta t, \quad (5)$$

where ΔI refers to the change in line current during the intra-hour stage; and the objective functions at the two stages are subjected to the following constraints.

$$v_{j,h} = v_{i,h} - 2(r_{ij,h} P_{ij,h} + x_{ij,h} Q_{ij,h}) + (r_{ij,h}^2 + x_{ij,h}^2) l_{ij,h}, \quad (6)$$

$$\sum_{k \in K(j)} P_{jk,h} = \sum_{i \in I(j)} (P_{ij,h} - r_{ij,h} l_{ij,h}) - P_{j,h}^l + P_{j,h}^{vsc}, \quad (7)$$

$$\sum_{k \in K(j)} Q_{jk,h} = \sum_{i \in I(j)} (Q_{ij,h} - x_{ij,h} l_{ij,h}) - Q_{j,h}^l + Q_{j,h}^{vsc}, \quad (8)$$

$$P_{ij}^2 + Q_{ij}^2 \leq l_{ij} v_i, \quad (9)$$

$$P_{g,h} = P_{01,h}; Q_{g,h} = Q_{01,h}, \quad (10)$$

$$P/Q_{ij,h}^{\min} \leq P/Q_{ij,h} \leq P/Q_{ij,h}^{\max}, \quad (11)$$

$$l_{ij,h}^{\min} \leq l_{ij,h} \leq l_{ij,h}^{\max}, \quad (12)$$

$$\left(V_{j,h}^{\min} \right)^2 \leq v_{j,h} \leq \left(V_{j,h}^{\max} \right)^2, \quad (13)$$

$$P_{j,h}^{vsc} = \left(P_{j,h}^{pv} - P_{c,j,h}^b + P_{d,j,h}^b \right) \eta_{vsc}, \quad (14)$$

$$Q^{vsc} = \sqrt{\left(\frac{3V_g X}{V_c} \right)^2 - P_{j,h}^{vsc2}} - \frac{3V_g^2}{X}, \quad (15)$$

$$P_{c,j,h}^b - P_{d,j,h}^b = \left(\pi_{c,j,h}^b + |\pi_{c,j,h}^b - \varphi_{c,j,h}^b| \right) - \left(\pi_{d,j,h}^b + |\pi_{d,j,h}^b - \varphi_{d,j,h}^b| \right), \quad (16)$$

$$P_{j,h}^{vsc} = \pi_{j,h}^{p,vsc} + \varphi_{j,h}^{p,vsc}, \quad (17)$$

$$Q_{j,h}^{vsc} = \pi_{j,h}^{q,vsc} + \varphi_{j,h}^{q,vsc}, \quad (18)$$

$$P_{j,h}^{pv} = \pi_{j,h}^{pv} + \varphi_{j,h}^{pv}, \quad (19)$$

$$\xi_{c/d}^{j,h} P_{d/c,j,h}^{b,\min} \leq \pi_{d/c,j,h}^b \leq \xi_{c/d}^{j,h} P_{d/c,j,h}^{b,\max}, \quad (20)$$

$$P_{j,h}^{vsc,\min} \leq \pi_{j,h}^{p,vsc} \leq P_{j,h}^{vsc,\max}, \quad (21)$$

$$Q_{j,h}^{vsc,\min} \leq \pi_{j,h}^{q,vsc} \leq Q_{j,h}^{vsc,\max}, \quad (22)$$

$$\xi_c^{j,h} + \xi_d^{j,h} \leq 1; \quad \zeta_c^{j,h} + \zeta_d^{j,h} \leq 1, \quad (23)$$

$$SOC^{t+\Delta t} = SOC^t + \frac{P_{c,j,h}^b \eta_b^c - P_{d,j,h}^b / \eta_b^d}{E_b} \Delta t, \quad (24)$$

$$SOC_{j,h}^{\min} \leq SOC_{j,h}^{t+\Delta t} \leq SOC_{j,h}^{\max}, \quad (25)$$

$$\xi_{c/d}^{j,h} P_{d/c,j,h}^{b,\min} \leq \varphi_{d/c,j,h}^b \leq \xi_{c/d}^{j,h} P_{d/c,j,h}^{b,\max}, \quad (26)$$

$$P_{j,h}^{vsc,\min} \leq \varphi_{j,h}^{p,vsc} \leq P_{j,h}^{vsc,\max}, \quad (27)$$

$$\Delta I_{ij,h} = \frac{(\varphi_{j,h}^{p,vsc} - \pi_{j,h}^{p,vsc}) + (\varphi_{j,h}^l - \pi_{j,h}^l)}{v_{j,h}} b_{ij}, \quad (28)$$

$$\sum_{j \in N_b} \sum_{h \in N_{ph}} \left(\varphi_{j,h}^l - \pi_{j,h}^l \right) - \left(\varphi_{j,h}^{p,vsc} - \pi_{j,h}^{p,vsc} \right) + r_{ij,h} \Delta I_{ij,h}^2 = 0, \quad (29)$$

where $r_{ij,h}$ is the resistance between i and j in phase h , K is the set of incoming branches and I refers to the set of outgoing branches at a node. P^l and Q^l are real and reactive loads; P^{vsc} and Q^{vsc} refer to real and reactive power outputs of the VSC; $\pi_{c/d}^b$ and $\psi_{c/d}$ are the battery's charging/discharging power and binary decision variable during hour-ahead operation; $\varphi_{c/d}^b$ and $\zeta_{c/d}$ are the battery's charging/discharging power and binary decision variables during intra-hour operation; E_b specifies the battery energy capacity and SOC refers to its state of charge; $\pi^{p,vsc}$ and $\varphi^{p,vsc}$ refer to the real power output of VSC in hour-ahead and intra-hour stages, respectively; π^l and φ^l are the real power load during hour-ahead and intra-hour stages, respectively, and b_{ij} refers to elements of the bus injection to branch current matrix.

Here, Equations (6–9) represent the branch power flow equations, Equation (10) are the line flows into the network from PCC, Equations (11) and (12) specify the line limits on real power, reactive power and current flows, Equation (13) specifies the nodal voltage limits, Equations (14) and (15) represent the real and reactive power outputs of the VSC, respectively, Equations (16–19) specify battery power, inverter output and PV output as time-decomposed signals during hour-ahead and intra-hour stages; Equations (20–27) specify the limits on battery power and VSC outputs in the two stages; Equation (23)

ensures single mode (charging/discharging) operation of the battery, Equations (24) and (25) refer to battery SOC and its limits, Equation (28) specifies the change in line current because of re-dispatch of BESS and VSC on occurrence of uncertainties, Equation (29) ensures balance of power deviations because of re-dispatch and uncertainties.

It is important to highlight that the network is assumed to be balanced, hence, a positive sequence equivalent modelling of the power system is used. The focus of this research is to develop the multi-timescale cyber-physical microgrid system as a first proof-of-concept. The challenge of unbalance in distribution systems would be addressed in future works. The challenges to incorporate unbalance in the current modelling are as follows:

- The developed optimisation framework uses the exact DistFlow model for modelling the radial distribution system, which is based on the full AC power flow equations, which are nonlinear [37]. This nonlinearity makes the model computationally challenging to incorporate phase unbalance and mutual coupling in the model.
- The laboratory hardware developed for our research setup, as depicted in Figure 6 of the manuscript, is a balanced three phase system. This aligns well with the use of the exact DistFlow model.

3.3 | Stochastic Programming Framework

The low-inertia power system presented as depicted in Figure 3, consists of several resources having distinct response times, control and ramping capabilities. For instance, the cost optimisation problem is based upon wholesale and retail price signals from the grid which are essentially hourly prices. On the contrary, batteries and voltage source converters have shorter response time and thus, can very efficiently be scheduled to track real-time power signals. Because of different response times and capabilities of the system components, the concept of timescale decomposition has been employed where the optimisation problem has been decomposed into two different levels to mitigate uncertainty occurrences in the network, that is, hourly and intra-hourly timescales. For the purpose of coordination among the two levels of optimisation, based on two-stage stochastic programming model was employed because of its capabilities to efficiently solve multi-stage problems with recourse decisions. The stochastic programming with recourse is essentially a two-stage problem which performs the first stage

optimisation in the absence of uncertainties and delays the second stage decision to a later time when uncertainties are actually present, or a precise information is available for the uncertain variables.

The proposed optimisation model presented in (4–29) is solved using the stochastic programming approach as a two-stage problem coordinated at different timescales. Accordingly, the longer timescale corresponds to hour-ahead optimisation that computes ‘wait-and-see’ decision variables x through the solution of $\min f(x) : x \in F$. Similarly, the shorter timescale corresponds to intra-hour optimisation at 15-min intervals wherein the ‘here-and-now’ optimal decision vector y is computed through the solution of the second stage problem $\min g(y) : y \in G(x, \psi)$, where ψ represents the occurrence of real-time uncertainties. Thus, the optimisation problem is reformulated as a two-stage stochastic programming model, expressed as follows [38].

$$\min\{f(x) + \mathbb{E}[H(x, \psi)]\}, \quad (30)$$

where $\mathbb{E}[H(x, \psi)]$ represent the expected values of intra-hour optimisation, respectively.

Uncertainties in the model are represented through scenario occurrences that are modelled using well-defined probability distribution functions (PDF). In particular, the stochastic variation in PV output and load are precisely modelled through beta and normal distributions, respectively. Further, the accuracy of uncertainty scenarios depends upon the number of scenarios considered for modelling the uncertainty and improves with increasing the number of samples. However, this increases the computational complexity as well as solution time. In view of this, suitable scenario reduction techniques have been widely used in literature, which reduce the computational burden while maintaining the accuracy of the representative scenario set. Appropriately, the backward reduction technique has been employed in this work to obtain an accurate scenario set to model the uncertainties.

To generate stochastic scenarios, the cumulative distribution function (CDF) of each uncertain source is computed and then Monte Carlo simulation is performed. However, the computational efficiency of the optimisation model depends on the number of scenarios generated. Thus, in order to minimise the computational complexity while simultaneously maintaining accuracy, the scenarios need to be reduced to a smaller number which can suitably represent the expected uncertainties. A number of scenario reduction techniques are available in

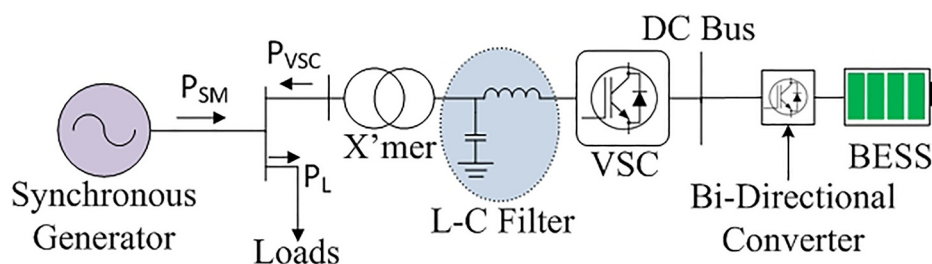


FIGURE 6 | Single line diagram of the experimental setup.

literature; however, the backward reduction technique was employed in this work.

Let ξ_n ($n = 1, \dots, K$) represent the initial generated scenarios, each with a probability λ_n . Let the distance between a scenario pair be $d_{n,n'}$ and $\bar{n}$ be the average value of scenarios. The distance can then be expressed as follows.

$$d_{n,n'} = \max\{1, \|n - \bar{n}\|, \|n' - \bar{n}\|\} \|n - n'\|. \quad (31)$$

For implementation of the backward reduction scheme, consider the initial set of scenarios and an empty set to store the deleted scenarios, represented by S and J , respectively. The steps involved in the scenario reduction scheme include the following steps.

1. Compute distances of all scenario pairs $d_{n,n'} = d(\xi_n, \xi_{n'})$ ($n, n' = 1, \dots, K$).
2. For each scenario k , $dt_k(r) = \min d_{k,n'}$. Here, $n', k \in S$, $n' \neq k$ and r is the scenario index that has the minimum distance to scenario k .
3. Determine $\lambda d_k(r) = \lambda_k \times dt_k(r)$. Choose d such that, $\lambda d_d = \min \lambda d_k$.
4. $S = S - \{d\}$, $J = J + \{d\}$ and $\lambda_r = \lambda_r + \lambda_d$.
5. Repeat steps (2)–(4) until the required number of scenarios R is obtained. Their respective probabilities are λ'_r , where $r = 1, \dots, R$.

Considering the occurrence of uncertainties ψ as a finite number of instances $\psi_1, \psi_2, \dots, \psi_k$, each associated with occurrence probabilities $\lambda_1, \lambda_2, \dots, \lambda_k$, respectively, the two-stage stochastic programming model is restructured to derive a deterministic equivalent, mathematically expressed as follows [38].

$$\mathbb{E}[H(x, \psi)] = \sum_{k=1}^K \lambda_k H(x, \psi_k). \quad (32)$$

$$\min_{x, y_1, \dots, y_k} f(x) + \sum_{k=1}^K \lambda_k q(y_k). \quad (33)$$

$$\text{s.t. } x \in F ; y_k \in G(x, \psi_k), \forall k. \quad (34)$$

For the proposed optimisation problem, the first stage objective $f(x)$ is given by Equation (4), subjected to the constraint set (6–25) and is computed hourly at $t = t_u$, where t_u refers to the hourly timescale. The optimal values of the first-stage decision variable x are applied hour-ahead, and are held constant until precise forecasts of stochastic uncertainties are available, that is, for $[t_u, t_u + \Delta t]$, where t_l refers to the intra-hourly time interval (15-min). However, during actual system operation, occurrence of system uncertainties ψ renders the optimal values of x sub-optimal. Accordingly, the second-stage optimisation presented in Equation (5) and subjected to Equations (23–29) is performed. Optimal operation decisions of the model are updated with the recourse decision variables y , and are applied for the duration of the intra-hour optimisation $t = [t_u + m\Delta t]$, $m = (1, 2, 3)$.

Whereas the accuracy of the generated scenarios is improved as the sample size of scenario set increases, it also significantly increases the solution time and computational complexity. Accordingly, the stochastic model of Equations (33–34) is reduced to a finite number of representative scenarios ($r = \{1, \dots, R\}$, where R is the set of reduced scenarios) using the backward reduction technique, to accurately represent the uncertainty while reducing the computational burden. Thus, the reduced stochastic optimisation model is mathematically represented as follows.

$$\min_{x, y_1, \dots, y_r} f(x) + \sum_{r=1}^R \lambda'_r q(y_r). \quad (35)$$

$$\text{s.t. } x \in F ; y_r \in G(x, \psi_r), \forall r. \quad (36)$$

3.4 | Real-Time Optimisation

Real-time events associated with large power switching may result in voltage and frequency disturbances in low-inertia networks. Conventional primary frequency control in low-inertia microgrids results in slower response and larger frequency deviations because of dynamic constraints of synchronous machines. In view of this, the proposed control methodology aims to achieve fast primary frequency regulation by providing transient power through power electronic converters. As a consequence, the real-time frequency fluctuations occurring in the network are minimised and frequency deviations are drastically reduced as compared to conventional control strategies. Accordingly, event-based real-time optimisation is performed as and when a switching decision is made by the building operator and the VSC is optimally re-dispatched. The real-time event is initiated when an appliance operation command is communicated by the building operator to the smart load control board. The real-time optimisation is performed by the central controller and is based on the concept of re-dispatching of powers by multiple VSCs in the network to meet the large power switching requirements so as to regulate primary frequency of the network while minimising line losses occurring because of real-time increase in line flows. The real-time optimisation model is expressed as follows.

$$\min \sum_{j \in N_b} \sum_{h \in N_{ph}} r_{ij,h} \Delta I_{ij,h}^{*2}, \quad (37)$$

subjected to

$$P_{j,h}^{vsc, \min} \leq P_{j,h}^{*p, vsc} \leq (P_{j,h}^{vsc, \max} - \phi_{j,h}^{p, vsc}), \quad (38)$$

$$SOC^{t*+\Delta t} = SOC^{t*} + \frac{P_{c,j,h}^{*b} \eta_b^c - P_{d,j,h}^{*b} / \eta_b^d}{E_b} \Delta t, \quad (39)$$

$$SOC_{j,h}^{\min} \leq SOC_{j,h}^{t*+\Delta t} \leq SOC_{j,h}^{\max}, \quad (40)$$

$$\zeta_{c/d}^{*j,h} P_{d/c,j,h}^{b, \min} \leq P_{d/c,j,h}^{*b} \leq \zeta_{c/d}^{*j,h} P_{d/c,j,h}^{b, \max}, \quad (41)$$

$$\zeta_c^{*j,h} + \zeta_d^{*j,h} \leq 1, \quad (42)$$

$$\Delta I_{ij,h}^* = \frac{(P_{j,h}^{*p,vsc} - \varphi_{j,h}^{p,vsc}) + (P_{j,h}^{*l} - \varphi_{j,h}^l)}{v_{j,h}^*} b_{ij}, \quad (43)$$

$$\sum_{j \in N_b} \sum_{h \in N_{ph}} (P_{j,h}^{*l} - \varphi_{j,h}^l) - (P_{j,h}^{*p,vsc} - \varphi_{j,h}^{p,vsc}) + r_{ij,h} \Delta I_{ij,h}^{*2} = 0, \quad (44)$$

where t^* denotes the real-time instant of load switching, $P^{*p,vsc}$ and P^{*b} refer to the real-time optimal power set-points of VSC and battery, respectively, P^{*l} refers to the power fluctuation associated with the real-time load switching and ΔI^* refers to change in current flows because of real-time re-dispatch of VSCs. Δt denotes the duration of primary frequency control. Here, Equation (38) represents the VSC capacity available in real-time for re-dispatch, whereas Equation (44) highlights that the real-time switched load and incremental losses must be balanced by VSC re-dispatch. The real time optimisation proposed in this work is an event-triggered optimisation that is performed as and when a microgrid's or building's central controller delegates a switching instruction. In accordance with the decision of the switching information, the optimisation is performed in real-time and the decision output is communicated for implementation to the local controller of the inverters. In this context, the accuracy of the solution depends on three major factors which are the solution algorithm, the generation and load uncertainties and the communication channel.

The real-time optimisation is a mixed integer quadratic programming problem, for which exact solution methods are well known. This paper performs the real-time optimisation through the Gurobi solver which is known to guarantee exact solutions for MIQP problems advanced solution algorithms such as branch-and-bound, branch-and-cut etc. [39].

In modern power systems that experience changing load compositions and greater penetration of renewable energy resources, load and generation are bound to exhibit stochastic behaviour. It should be noted that the proposed real-time optimisation control inverter operation in response to any scheduled switching in the system. The proposed control does not account for real-time uncertainty occurrences that arise in the network. However, these uncertainties are effectively mitigated in this work through load frequency control by the low-inertia power system.

The communication channel is the main determinant in the accuracy of the proposed real-time optimisation algorithm because the optimisation relies upon communication of the switching decision from the building operator to the central controller, and from the central controller to the local inverter controller. However, this work involves communication over shorter distances, which minimises the risk of loss of communication channel. Moreover, this work uses WiFi as the communication channel, which is typically regulated by communication standards and has the mechanism to detect and correct distortion and disturbances to ensure data integrity.

ALGORITHM 1 | Coordination framework for multi-timescale optimization and real-time control.

```

1 Hour-ahead operation
2 Inputs:  $\pi^{pv}$ ,  $\pi^l$ ,  $SOC^t$ 
3 while  $t_u \in T$ , where  $T = [1, 24]$  do
4   Optimize (4) subjected to (6)-(16), (20)-(25)
5   Control signal  $P_{ref} \rightarrow VSC$ 
6    $P_{ref} = \pi^{p,vsc}$  &  $Q_{ref} = \pi^{q,vsc}$ 
7   Intra-hour operation
8   Inputs:  $\varphi^{pv}$ ,  $\varphi^l$ ,  $SOC^t$ ,  $v$ ,  $\pi^{p,vsc}$ ,  $\pi^{q,vsc}$ 
9   while  $t = (t_u + m\Delta t)$ ,  $\forall m = (1, 2, 3)$  do
10    Stochastic modelling of  $\varphi^{pv}$  and  $\varphi^l$ 
11    Optimize (5) subjected to (23)-(29)
12    Updated control signal  $P_{ref} \rightarrow VSC$ 
13     $P_{ref} = \varphi^{p,vsc}$ 
14    Real-time operation ( $t^*$ )
15    Switching command  $\rightarrow$  Central controller
16    Inputs:  $P^{*l}$ ,  $\varphi^{p,vsc}$ ,  $SOC^{*t}$ ,  $v^*$ 
17    Optimize (37) subjected to (38)-(44)
18    Real-time control signal  $\Delta P_{VSC} \rightarrow VSC$ 
19     $\Delta P_{VSC} = (P_{j,h}^{*p,vsc} - \varphi_{j,h}^{p,vsc})$ 
20    At  $t = t^* + \Delta t'$ 
21    Restore VSC control  $P_{ref} = \varphi^{p,vsc}$ 
22    Active power recovery by grid ( $P_g$ )

```

3.5 | Real-Time Control Framework

Optimal control outputs from time-ahead and real-time optimisation are communicated to VSC which further controls BESS operation. Coordination between the time-ahead and real-time optimisation, and control is illustrated in Algorithm 1. Optimal VSC dispatch decisions in hour-ahead, intra-hour and real-time timescales, that is, $\pi^{p/q,vsc}$, φ^{vsc} and P^{*vsc} , respectively, are commands for real-time control of VSC and are communicated as described in the algorithm.

The VSC active and reactive power control diagram for BESS is shown in Figure 7. The VSC connected to BESS operates in grid following mode, the line voltages are sensed and passed through voltage balancing circuit using second-order generalised integrator (SOGI) which reduces the noise or harmonics or voltage unbalance in the system [40]. V_a , V_b and V_c are estimated balanced phase voltages and the amplitude (V_s) is estimated as follows.

$$V_s = \sqrt{\frac{2}{3}(V_a^2 + V_b^2 + V_c^2)}. \quad (45)$$

The estimated phase voltages (V_a , V_b and V_c) are passed through SOGI-PLL, which estimates grid voltage angle (θ). The estimated phase voltages (V_a , V_b and V_c) and (θ) are utilised to transform voltages in their corresponding $d - q$ reference frame, that is, V_d and V_q and similarly VSC line currents (I_a , I_b and I_c) are transformed to I_d and I_q . The active and reactive power of VSC, that is, P_{vsc} and Q_{vsc} , are calculated as follows.

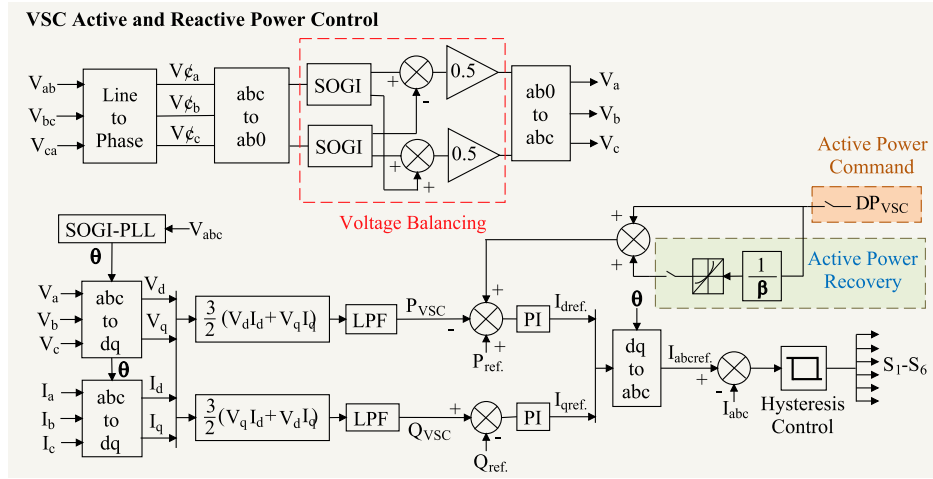


FIGURE 7 | Control diagram of VSC.

$$P_{vsc} = \left[\frac{3}{2} (V_d I_d + V_q I_q) \right] \left(\frac{2\pi f_{cf}}{s + 2\pi f_{cf}} \right). \quad (46)$$

$$Q_{vsc} = \left[\frac{3}{2} (V_q I_d - V_d I_q) \right] \left(\frac{2\pi f_{cf}}{s + 2\pi f_{cf}} \right). \quad (47)$$

The VSC reference current in the rotating ($d - q$) frame is calculated as follows.

$$I_{dref} = \left(K_{pv} + \frac{K_{iv}}{s} \right) \left[P_{ref} - P_{vsc} + \Delta P_{vsc} \left(1 + \frac{1}{\beta} \right) \right], \quad (48)$$

$$I_{qref} = \left(K_{pv} + \frac{K_{iv}}{s} \right) [Q_{vsc} - Q_{ref}], \quad (49)$$

where P_{ref} , Q_{ref} and ΔP_{vsc} are obtained from the multi-timescale optimisation as discussed in Algorithm 1. Furthermore, a two-stage converter control strategy has been considered for BESS. The BESS is connected to DC bus through bi-directional DC-DC converter, which regulates the DC bus voltage of VSC using voltage-current control loops. The control signals to bi-directional DC-DC converter for DC bus voltage control are generated using the following.

$$\Delta V_{DC} = V_{DCref} - V_{DC} \left(\frac{2\pi f_{cf}}{s + 2\pi f_{cf}} \right), \quad (50)$$

$$I_{Bref}(t) = k_{pdc} \Delta V_{DC} + k_{idc} \int \Delta V_{DC} \cdot dt, \quad (51)$$

$$\Delta I_B = I_{Bref} - I_B \left(\frac{2\pi f_{cf}}{s + 2\pi f_{cf}} \right), \quad (52)$$

$$D_B = k_{pb} \Delta I_B + k_{ib} \int \Delta I_B \cdot dt, \quad (53)$$

where V_{DC} is the actual DC bus voltage which is passed through low-pass filter, whose cut-off frequency (f_{cf}) is chosen as 5 Hz [41]. V_{DCref} is the reference DC bus voltage, ΔV_{DC} is the change in DC bus voltage. k_{pdc} and k_{idc} are the proportional and integral gain

constant of outer voltage control loop. I_{Bref} is the reference battery current calculated from outer voltage control loop and I_B is the actual battery current. ΔI_B is the difference in reference and actual battery current. k_{pb} and k_{ib} are the proportional and integral gain constant of inner current control loop. D_B is the duty cycle for bi-directional DC-DC converter which is converted to pulse width modulating signals with switching frequency of 5 kHz.

Similarly, a two-stage converter control strategy is considered for solar PV generation. The solar PV array is connected to DC-DC boost converter which tracks the maximum power point (MPP) of solar PV using the incremental conductance control (INC) MPPT algorithm. The output of DC-DC boost converter is interlinked to AC bus through voltage source converter (VSC). The VSC regulates the DC bus voltage and operates on grid following mode by following the reference of grid frequency.

When the frequency is restored within permissible limits after the disturbance, the VSC power needs to be restored to its optimal dispatch. The active power recovery method slowly restores back the VSC power to its original dispatch without disturbing system frequency (within prescribed limits). The power recovery is done by multiplying the negative slope to ΔP_{vsc} . The value of slope is calculated based on the active power disturbance and dynamics of the grid. The slope maintains the change in active power such that the demanded power is supplied by the grid.

4 | Simulation Results

The proposed multi-timescale optimisation and control approach was simulated on the modified CIGRE LV European test system, as depicted in Figure 8. A solar PV generation system of 4 kW is connected at bus no. 16, whereas BESS is connected at bus nos. 15 and 10. Bus voltage at the root bus is 1.0 pu, whereas minimum and maximum voltage limits are considered to be 0.9 and 1.1 pu, respectively. Battery charging and discharging efficiencies are considered to be 97% each, and SOC is limited between 0.25 and 1. Battery decay rate is considered to be 0.001, whereas VSC operation is limited by the converter power capability curve.

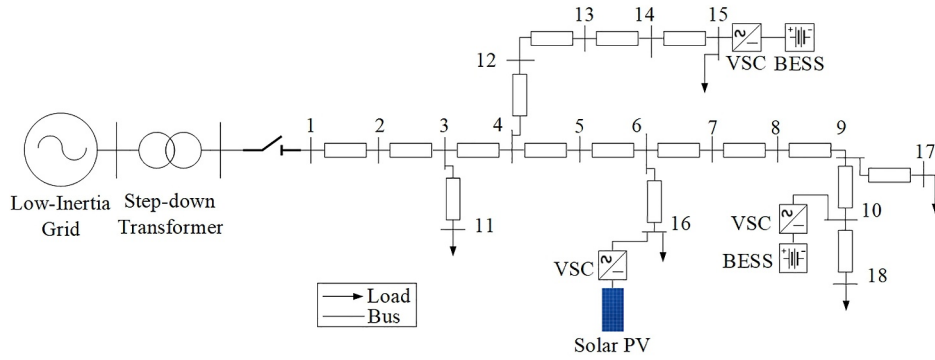


FIGURE 8 | Modified CIGRE European LV distribution network [42].

TABLE 1 | Comparative analysis of deterministic and two-stage stochastic optimisation (all values in kWh).

Case	E^l	E^{pv}	E^{vsc}	E_d^b	E_c^b	E^{loss}
Deterministic	13.87	3.90	6.24	4.92	1.88	0.47
Stochastic	12.03	3.78	4.40	4.39	3.28	0.48

A deterministic test case has been simulated considering the absence of real-time uncertainties in load and PV output, and the results obtained have been depicted in Table 1. Furthermore, to analyse the dynamic performance of the deterministic solution, the single-point predictions of load at bus 11 and solar PV at bus 16 are compared against 1000 uncertainty samples generated using the Monte Carlo simulation. The comparison illustrated in Figure 9 depicts that actual occurrence of PV output and load may vary significantly in real time, rendering the deterministic solutions as sub-optimal for system operation.

The proposed two-stage stochastic optimisation strategy mitigates the effects of real-time uncertainties through probabilistic modelling of stochastic uncertainties. In view of this, load and PV uncertainties are modelled as normal and beta PDFs, respectively. Accordingly, 1000 uncertainty scenarios are generated using the Monte Carlo sampling technique. The backward reduction technique was implemented to reduce the large number of uncertain scenarios to a smaller representative set ($R = 30$). The two-stage stochastic optimisation was then implemented for the reduced scenario set and the results obtained have been depicted in Table 1 and Figures 10 and 11. It can be observed from Figure 11 that because of stochastic modelling of uncertainties with updated forecasts in 15-min, the values of load and PV output vary intra-hourly, in contrast to the deterministic forecasts. As a consequence, the charging/discharging of BESS and VSC power output is updated every 15-min to meet the intra-hourly uncertainties. The results for simulation over an hour are tabulated in Table 1 and depict that because of stochastic uncertainties, the hourly expectation (energy) of load (E^l) and PV power output (E^{pv}) are reduced by 13% and 3%, respectively, because of which battery charging energy (E_c^b) is increased from 1.88 to 3.28 kWh while discharging energy (E_d^b) is decreased by 0.53 kWh for approximately the same line losses. As a result, VSC energy output (E^{vsc}) over 1 h reduces from 6.24 to 4.40 kWh.

Optimal dispatch outputs from stochastic optimisation are given as inputs to Simulink model of the modified Cigre LV network and results for an intra-hourly instant are illustrated in Figures 12–15. The hour-ahead power dispatch of VSCs at bus 10, 15 and 16 are shown in Figure 12. As depicted in the figure, at the instant when PV is generating 1.79 kW, the BESS VSC at bus 10 is generating an active power of 1.42 kW and BESS VSC at bus 15 is generating 3 kW, as per the deterministic values generated from hour-ahead optimisation. However, because of the occurrence of uncertainties in smaller timescales, as depicted at 3 s in the figure, a recourse decision has to be taken using the second stage of the stochastic optimisation model. Because of the uncertainty occurrence, PV VSC output falls to 0.72 kW as shown in Figure 13. The recourse optimisation is implemented and VSC are re-dispatched to meet the uncertainty in PV output. Accordingly, BESS VSC at bus 10 is re-dispatched to 1.85 kW, whereas the BESS VSC at bus 15 is re-dispatched to 2 kW. Figure 14 shows the intra-hour load uncertainty. After the occurrence of load uncertainty at time $t = 3$ s, the loads at bus 17 and 15 are updated to 4.24 kW. Similarly the load at buses 11, 16 and 18 are changed to 2.5 kW, 1.3 and 1.4 kW, respectively. It is observed that when an uncertain load change occurs at an intra-hourly interval, the hour-ahead deterministic dispatch of VSCs results in a higher frequency deviation and longer settling time as depicted in Figure 15. Updated VSC dispatch obtained from stochastic optimisation are shown in Figure 13, which result in significantly faster frequency restoration as depicted in Figure 15.

Further, real-time optimisation and control results are illustrated in Figures 16–18. It is observed that when a real-time load of 1 kW is switched on at bus 11, the stochastic dispatch for VSC power is no longer optimal and the real-time frequency changes as shown in Figure 16. Thus, when a switching action of 1 kW is planned at bus 11, the building operator communicates the switching decision to the cloud database. Consequently, the central controller computes and communicates the optimal real-time VSC power control decisions for implementation at the load switching instant, as shown in Figure 17. Accordingly, the VSC power dispatch at bus 10 and 15 are increased by 658 and 343 W, respectively, to meet the increased load and incremental line losses. Correspondingly, because of the pre-disturbance optimal corrective action, the real-time frequency is controlled through the combined action of the VSCs and grid as shown in Figure 16. As the load switching action is implemented, the grid responds through its inertial response, followed by optimal real-

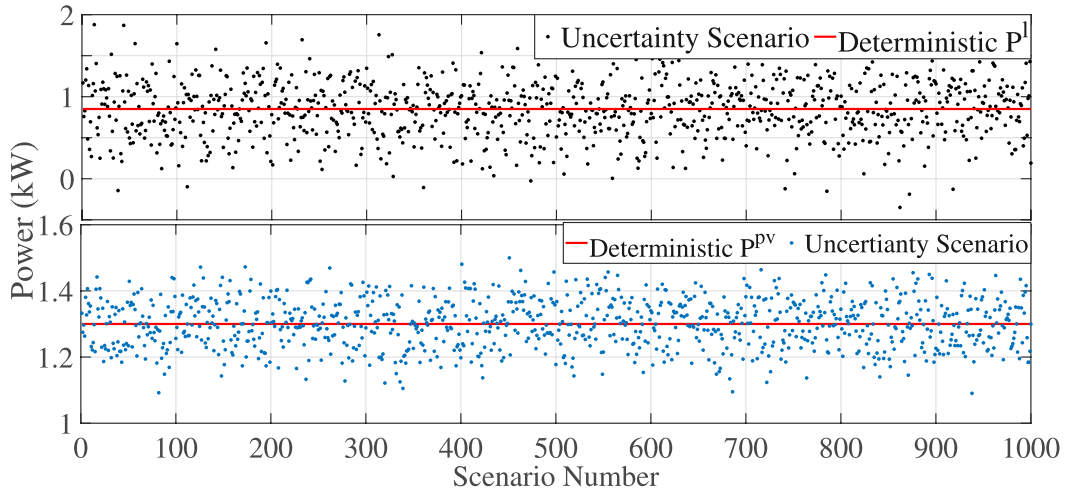


FIGURE 9 | Uncertainty scenarios for P^l and P^{pv} .

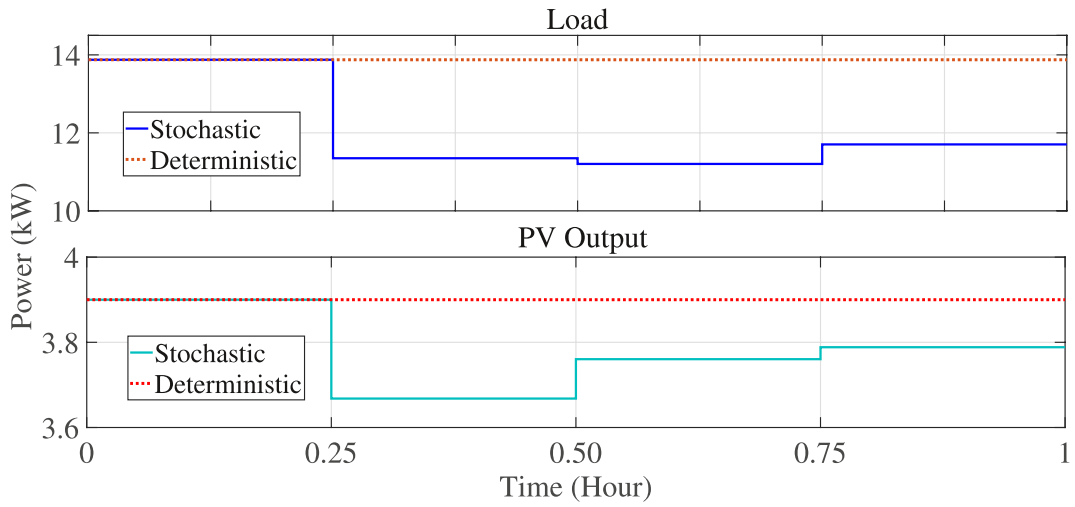


FIGURE 10 | Expected load and PV output.

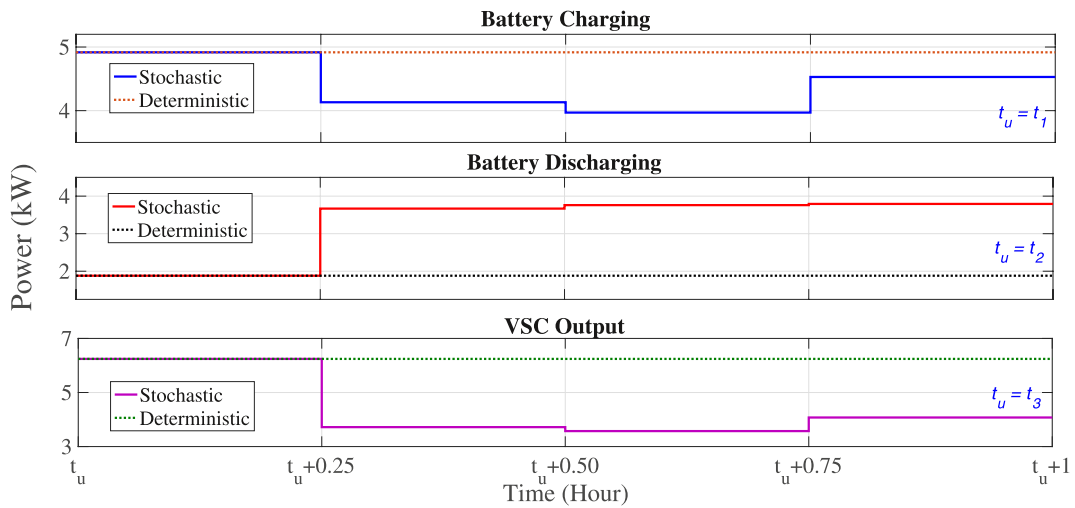


FIGURE 11 | Stochastic dispatch of BESS and VSC.

time VSC's power dispatch to control the primary frequency. Once the primary frequency is controlled by supplying active power through the VSCs, the VSC power operation is gradually

shifted to their stochastic dispatch set-points and the active load is gradually shifted to the grid. This is referred to as active power recovery control and is illustrated in Figure 18.

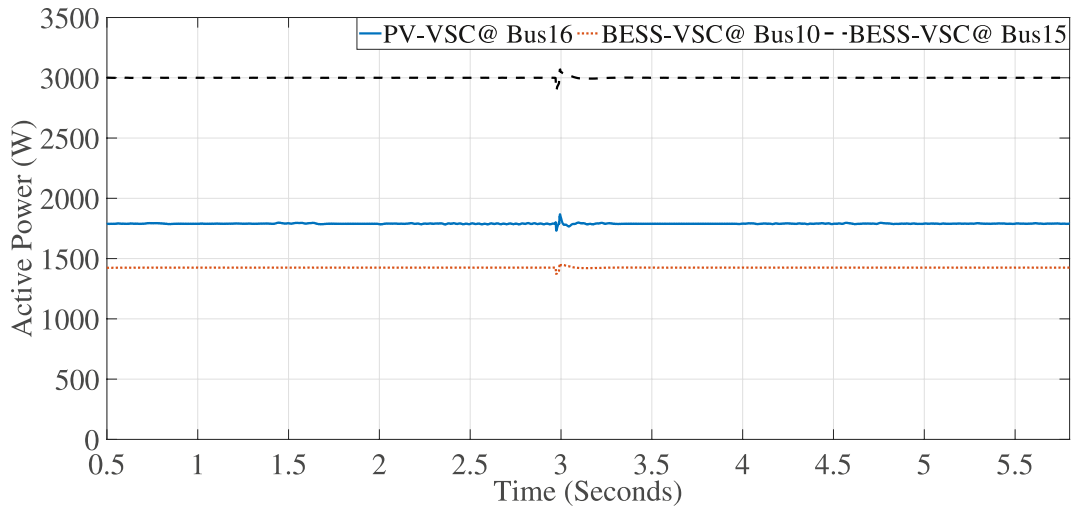


FIGURE 12 | Deterministic dispatch of VSC.

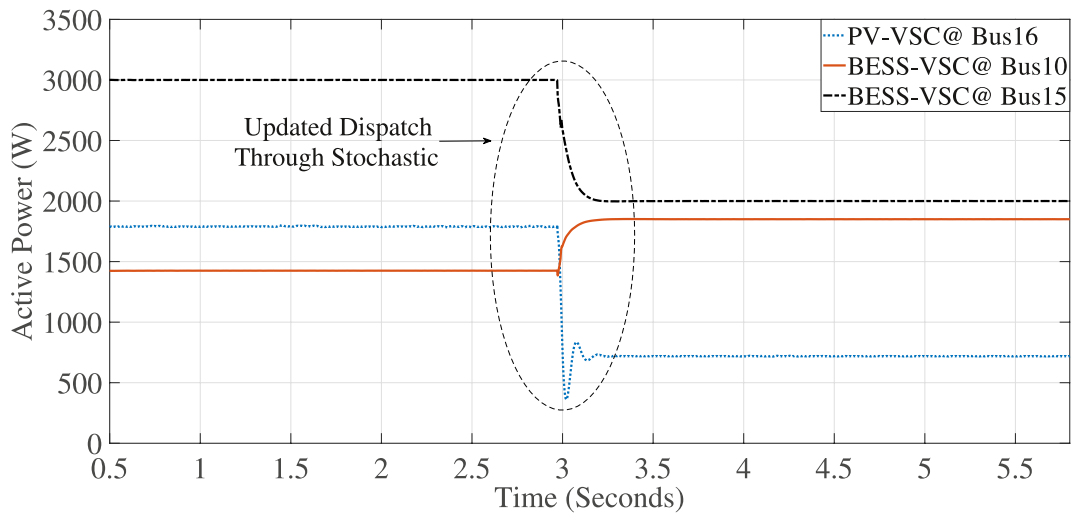


FIGURE 13 | Intra-hourly updated VSC output.

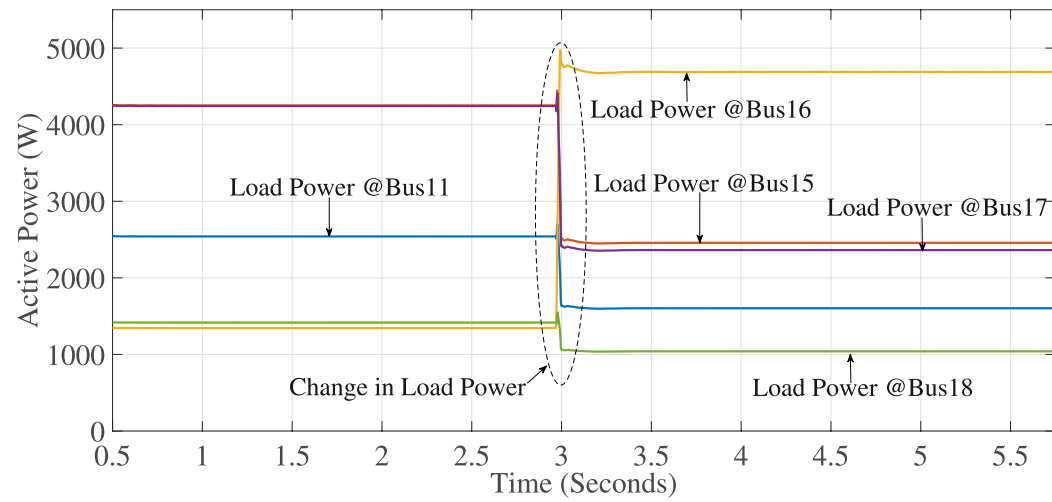


FIGURE 14 | Intra-hourly load uncertainty.

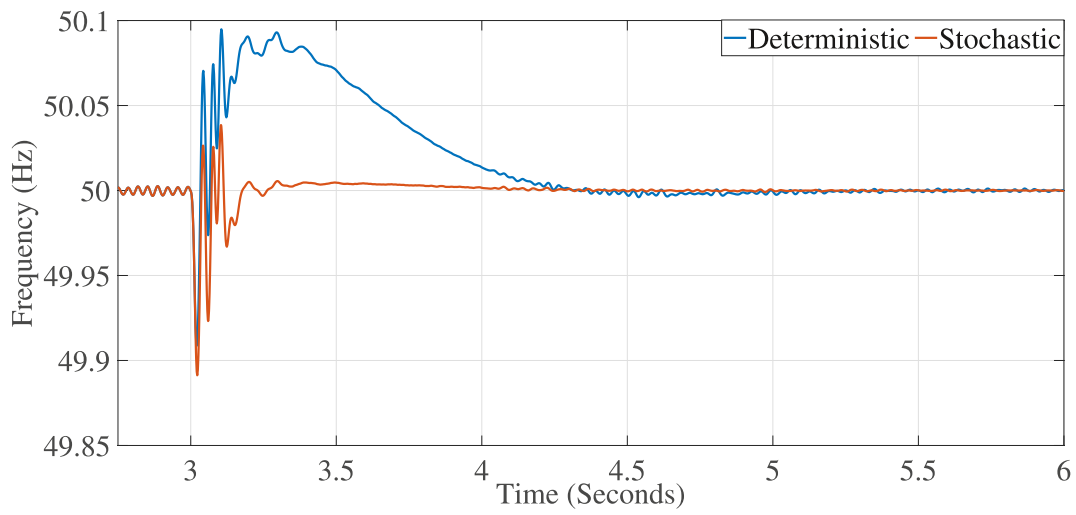


FIGURE 15 | Frequency with stochastic dispatch.

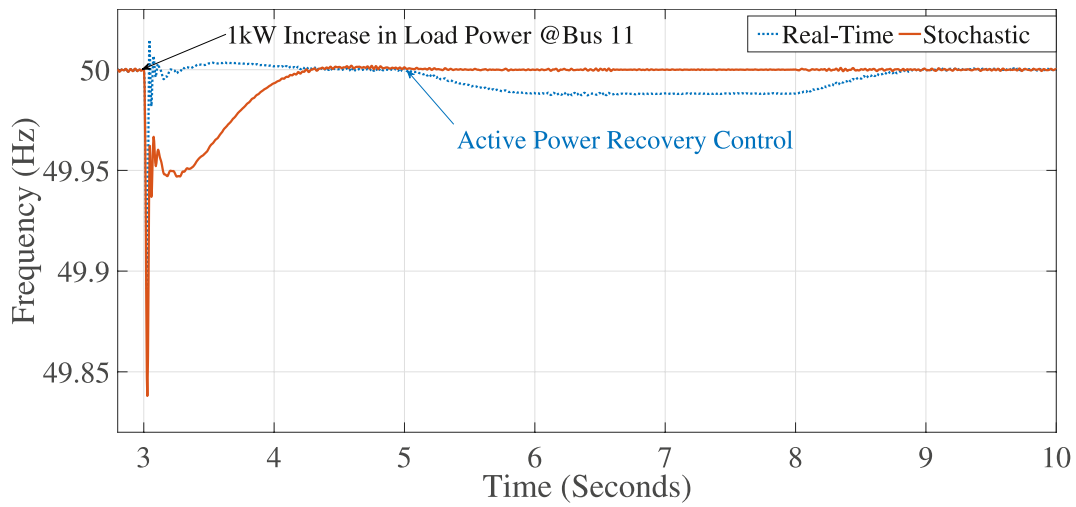


FIGURE 16 | Frequency with real-time optimisation.

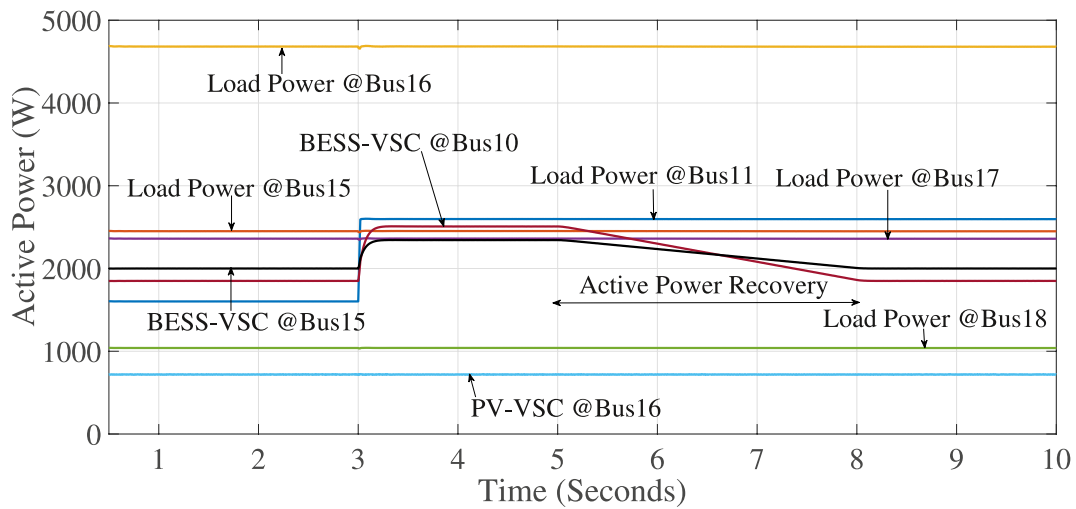


FIGURE 17 | Real-time load switching.

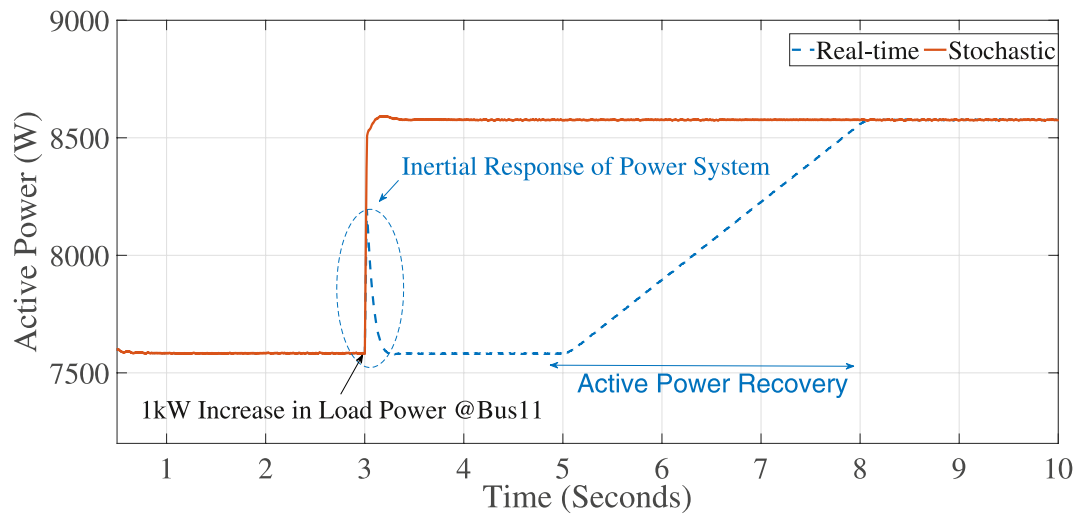


FIGURE 18 | Active power recovery.



FIGURE 19 | Lab-scale experimental setup.

5 | Experimental Results

5.1 | Experimental Test System

To validate the effectiveness of the proposed control strategy for reduction of frequency deviation, a laboratory scale experimental setup was developed as shown in Figure 19 and detailed diagram of the laboratory scale setup is shown in Figures 20. The single-line diagram of the developed microgrid setup is

shown in Figure 6. As shown in the figures, the setup consists of a DC motor-driven synchronous machine which mimics the behaviour of a low-inertia power system. The load frequency is regulated by rotating the shaft of synchronous machine at rated synchronous speed, which is indirectly controlled by controlling the mechanical power through DC motor. The mechanical power is controlled by keeping the field winding of DC motor at fixed voltage and regulating the armature voltage of DC motor through DC-DC buck converter.

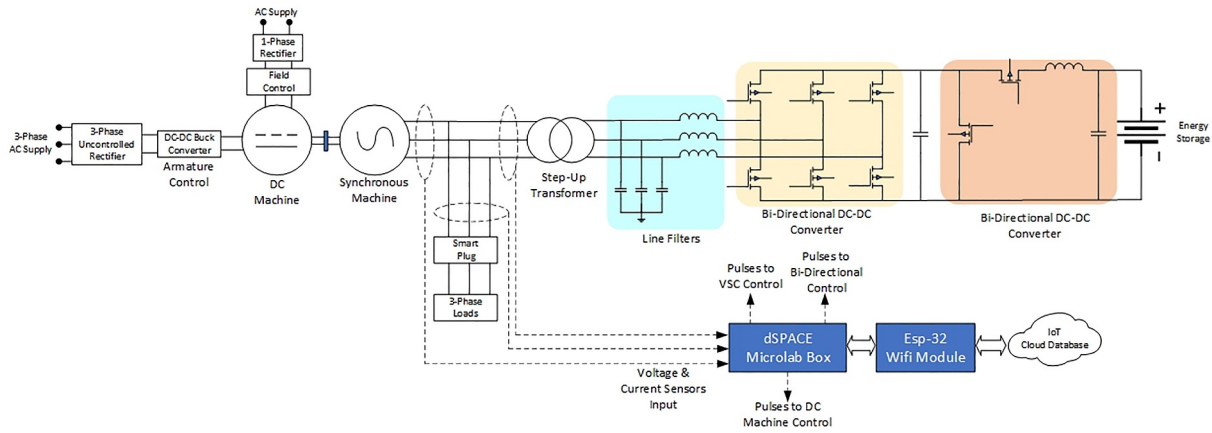


FIGURE 20 | Detailed experimental setup.

Power is fed to the DC–DC buck converter through a 12-pulse uncontrolled rectifier by real-time control of duty ratio using PI controller. Furthermore, the load voltage is regulated by controlling the duty ratio of the DC–DC buck converter using integral controller. The DC–DC buck converter is connected to the field winding of synchronous machine through a single phase uncontrolled rectifier. The setup consists of two different sources which are synchronous machine low-inertia power system and a battery-connected VSC, along with resistive loads. The battery is interfaced to the DC bus of VSC through the bi-directional DC–DC converter, which regulates the DC link voltage of VSC. VSC controls the active power and reactive power flow to microgrid by operating in grid following mode and follows the frequency and voltage regulated by synchronous machine. Conventionally, the battery operated VSC operates in constant power generation mode which is optimally calculated through various control methods and any mismatch in power because of load uncertainty is managed by the grid through inertial, primary response etc.

The overall communication framework of the proposed primary frequency control system is shown in Figure 21. As depicted in the figure, the user sends an appliance on/off command signal to Google firebase cloud platform service through a mobile/web application. The command signal updated on the cloud platform is retrieved through ESP32 WIFI module using IEEE 802.11 protocol. ESP32 module sends the command to dSPACE microlab box through analogue signals. The dSPACE microlab box acts as central controller, acquires the voltage and current sensor inputs and generates pulses at a sampling rate of 40 μ S. The central controller controls the appliance by sending on/off command to the smart plug as requested by the user. The controller modelling and tuning were performed in MATLAB/Simulink and communicated to central controller through control desk using wired LAN communication.

The conventional load frequency control uses PI controller to regulate load frequency deviation occurred due to active power mismatch between generation and load. This section handles the plant model identification and PI controller gain tuning of load frequency control. The overall plant model transfer function is identified by logging the duty ratio values given to the DC–DC buck converter for armature current control and the

output frequency generated by the synchronous machine. The logged data is fed to MATLAB's system identification tool and a second order transfer function is acquired with an accuracy of 98.30%. The overall plant transfer function is identified as follows [1].

$$G_p(s) = \frac{2.68e5}{s^2 + 303.4s + 4661}.$$

Because the plant transfer function is a second-order system, the poles lies on the left side on s-plane, therefore it is stable. The PI controller gain parameters are calculated using the Routh–Hurwitz stability criterion method. From the root-locus plot of closed loop system with PI controller, it is observed that the poles lies on the left side of s-plane, which verifies the stability criterion of the overall system with feedback control.

The system parameters considered in experimental setup is depicted in Table 2.

5.2 | Test Case

A test case is performed to examine the effect of low-inertia microgrid during any load disturbance. In this case, a synchronous machine is operated at unity power factor by providing 1.72 kW active power and VSC is providing 0.5 kW of active power to 2.22 kW resistive load. A command signal is sent to turn on 0.63 kW of appliance, such a step load change causes frequency deviation of 0.7 Hz as shown in Figure 22. The active power imbalance between load and source is provided by synchronous machine, as synchronous machine based source is made responsible for regulating the system load frequency. An increase of 0.63 kW is provided by the synchronous machine which regulates the frequency in 2.5 s and no change in change in VSC power is observed as shown in the figure.

Another test case is performed to examine performance of the proposed primary frequency control strategy. Similar to the previous case, synchronous machine is operated at unity power factor by providing 1.72 kW active power, whereas VSC provides 0.5 kW active power to resistive load. A command signal is sent to turn on an appliance of 0.63 kW. The proposed control

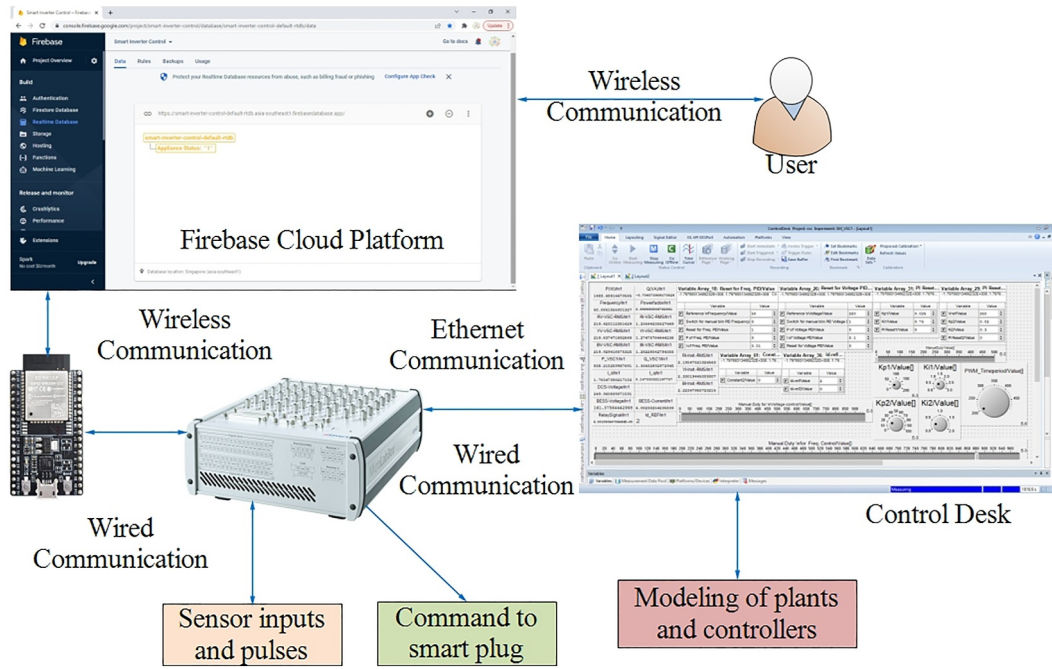


FIGURE 21 | Overall communication framework.

TABLE 2 | Experimental setup system parameters.

System parameters
<i>Synchronous machine:</i> 3 kVA, 3 – ϕ , 415V _{l-lrms} 50 Hz, 4-pole, 1500 rpm
<i>DC machine:</i> 5 H.P., 1500 rpm, 230 V, 21.5 A, shunt wound type
<i>BESS:</i> Lead acid type, no. of batteries: 14 nos, size of each battery: 12 V, 65 Ah
<i>VSC:</i> 3 kVA, 415V _{l-lrms} , $L = 6$ mH, $C = 25$ μ F, $f_{cf} = 5$ kHz. DC link voltage: 250 V, DC capacitor: 4500 μ F, 450 V
<i>Load:</i> 3 – ϕ resistive type, 2.25 kW, 415V _{l-lrms}
<i>Bi-directional DC/DC converter:</i> $L = 1.6$ mH, $C_{Bdc} = 1500$ μ F

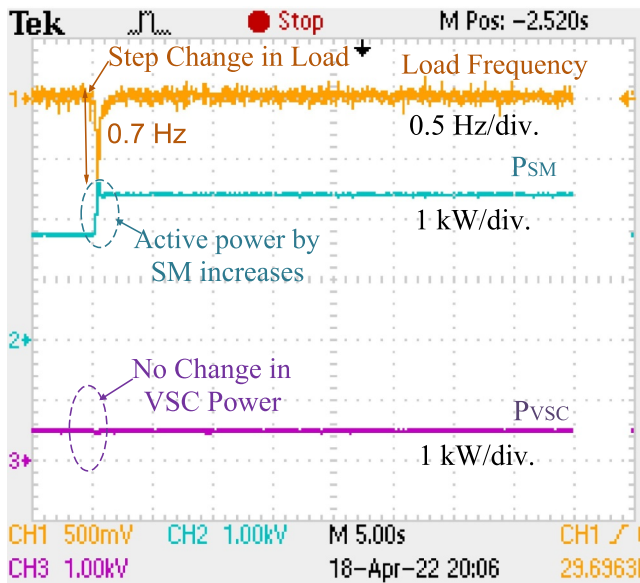


FIGURE 22 | Dynamic response with a step load change under conventional PI control.

strategy utilises this information for regulating frequency. Because VSC is operating at constant active power dispatch, because of the appliance turning on, a step change of 0.63 kW is added to VSC active power dispatch. It can be observed from Figure 23 that peak frequency undershoot is reduced to 0.1 Hz for same step change in load. Thus, VSC provides 1.14 kW active power for 10 s, after which active power recovery control mode is initiated. The active power recovery control restores the active power of VSC to its previous set point of 0.5 kW. During active power recovery mode, VSC power is complimented by power from synchronous machine to meet the desired load. Accordingly, while VSC power reduces from 1.14 to 0.5 kW, active power from synchronous machine simultaneously increases from 1.72 to 2.36 kW, thus maintaining the load frequency within the acceptable limits as shown in the figure.

6 | Conclusion

This paper presents a multi-timescale optimal control strategy for low-inertia electrical microgrid systems with high RES

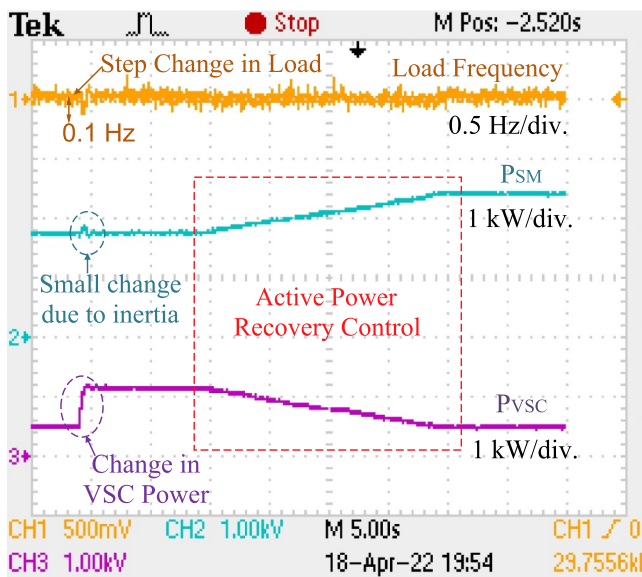


FIGURE 23 | Dynamic response with a step load change with proposed control.

penetration through adaptive VSC power control for fast frequency and voltage control. Accordingly, a two-stage stochastic optimisation model has been developed for adaptive dispatch of power from BESS and VSCs in hourly and intra-hour timescales, to counteract the effect of PV and load uncertainties on system stability. Furthermore, in view of the appliance switching/scheduling information available in smart grid environment, a real-time optimisation and control framework was developed for pre-disturbance preventive control of network frequency. Accordingly, a novel real-time coordination framework has been developed for primary frequency regulation by adaptive VSC power control and recovery control through grid. Effectiveness of proposed multi-timescale coordinated control is verified through extensive simulations and lab-scale experimental setup. Experimental results validate the performance of proposed microgrid system under study for real-time optimisation and control scheme for fast frequency regulation under large real-time load switching.

7 | Future Work

This Section discusses the major pathways identified from this research as future research. The first factor to adopt in future research is to incorporate distribution system unbalance in the studies. Accordingly, future research would adopt suitable linearisation approaches to incorporate multi-phase modelling of the distribution power network that would enable accurate representation of phase unbalancing. Further, the lab-scale hardware setup would be extended to emulate physical system unbalance.

Further, although the current analysis provides valuable insights through simulation-based case studies, the generalisability of the findings would benefit from validation in larger and more diverse power systems. Future work will focus on conducting comprehensive case studies involving a wider range of grid configurations, including both urban and rural networks,

and considering different levels of renewable energy penetration. Real-world hardware-in-the-loop (HIL) experiments and field demonstrations will also be pursued to assess the effectiveness of pre-event VSC dispatch strategies under practical operating conditions and real disturbances.

Additionally, further investigation would be performed to evaluate the impact of pre-event VSC dispatch under various contingency scenarios, such as faults, sudden load changes and renewable generation variability. This will involve developing more advanced optimisation frameworks that can account for system uncertainties and dynamic responses in real time. Finally, future research will explore the integration of data-driven and adaptive control methods to enable more robust and flexible pre-event dispatch strategies. By leveraging machine learning and real-time monitoring, it may be possible to further enhance the resilience and adaptability of microgrid control systems.

Nomenclature

Abbreviations

BESS	Battery energy storage system
DER	Distributed energy resources
E_c^b	Battery charging energy
E_d^b	Battery discharging energy
E^l	Hourly expectation (energy) of load
E^{pv}	Hourly expectation (energy) of PV power output
E^{vsc}	VSC energy output
INC	Incremental conductance control
IoT	Internet-of-things
LQR	Linear-quadratic regulator
LV	Low-voltage
MPC	Model predictive control
MPP	Maximum power point
MQTT	Message queuing telemetry Transmission
PCC	Point-of-common coupling
PV	Photo-voltaic
RES	Renewable energy sources
RNN	Recurrent neural network
RoCoF	Rate of change of frequency
SAA	Simulated annealing algorithm
SOGI	Second-order generalised integrator
ULFS	Under-frequency load shedding
VSC	Voltage source converter

Parameters

δ	Angle between grid and converter voltage
ΔI	Change in line current during the intra-hour stage
ΔI_b	Difference in reference and actual battery current
ΔV_{DC}	Change in DC bus voltage

$\mathbb{E}[H(x, \psi)]$	Expected values of intra-hour optimisation
ψ	Occurrence of real-time uncertainties
θ	Estimated grid voltage angle
$f(x)$	First stage objective
I	Set of outgoing branches at a node
K	Set of incoming branches
P	Active power
Q	Reactive power
R	Set of reduced scenarios
S	Apparent power
SOC	State of charge
X	Equivalent line reactance
$\eta_{b/d}^{d/c}$	Discharging/charging efficiencies of the battery
π^l	Real power load during hour-ahead stages
$\pi^{p,vsc}$	Real power output of VSC in hour-ahead
$\pi^{p/q,vsc}$	Command for real-time control of VSC
$\pi_{c/d}^b$	Battery's charging/discharging power during the hour-ahead operation
$\psi_{c/d}$	Battery's binary decision variable during the hour-ahead operation
φ^l	Real power load during intra-hour stages
$\varphi^{p,vsc}$	Real power output of VSC in intra-hour stages
φ^{vsc}	Command for real-time control of VSC
$\varphi_{c/d}^b$	Battery's charging/discharging power during the intra-hour operation
$\zeta_{c/d}$	Battery's binary decision variable during the intra-hour operation
b_{ij}	Elements of the bus injection to branch current matrix
C_b	BESS operational cost
C_g	Unit grid cost
D_B	Duty cycle for bi-directional DC–DC converter
E_b	Battery energy capacity
f_{cf}	Cut-off frequency
I_a	VSC line current of phase a
I_{Bref}	Reference battery current
I_b	VSC line current of phase b
I_B	Actual battery current
I_c	VSC line current of phase c
k_{ib}	Integral gain constant of inner current control loop
k_{idc}	Integral gain constant of outer voltage control loop
k_{pb}	Proportional gain constant of inner current control loop
k_{pdc}	Proportional gain constant of outer voltage control loop
L_f	Filter inductance
L_g	Line inductance
N_b	Sets of buses
N_{ph}	Sets of phases
P^{*vsc}	Command for real-time control of VSC
P^l	Real loads

P^{vscs}	Real power output of VSC
$P_{d/c}^b$	Discharging/charging power of BESS
P_g	Power drawn from the grid
P_{vsc}	Active power of VSC
Q^l	Reactive loads
Q^{vsc}	Reactive power of VSC
Q_{vsc}	Reactive power of VSC
$r_{ij,h}$	Resistance between i and j in phase h
t_l	Intra-hourly time interval
t_u	Hourly timescale
V_a	Estimated balanced voltage of phase a
V_b	Estimated balanced voltage of phase b
V_c	Estimated balanced voltage of phase c
V_c	Inverter voltage
V_{DCref}	Reference DC bus voltage
V_{DC}	Actual DC bus voltage
V_d	Transformed voltage in d -frame
V_g	Grid voltage
V_q	Transformed voltage in q -frame
V_s	Estimated voltage amplitude

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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